

Step-by-Step DSP Implementation & Synthesis Guide

DESIGN WORKFLOWS, INTERACTIVE BODE ANALYSIS, AND EMBEDDED C/C++ CODE GENERATION

1. Objective & Pedagogical Goals

This laboratory tutorial provides a comprehensive guide for synthesizing, analyzing, and deploying digital filters using Overtune 3 in conjunction with the EE 603 curriculum. Students will translate continuous frequency specifications into discrete filter prototypes, verify system stability in the complex z -plane, and export production C/C++ Second-Order Section routines for real-time DSP hardware.

2. Filter Synthesis Workflow

Step 1: Formulate Boundary Specifications

Define the target sampling rate f_s and boundary frequencies:

- **Sampling Rate (f_s):** Standard rates include 44.1 kHz (audio CD), 48.0 kHz (professional studio), and 96.0 kHz (high-resolution audio).
- **Passband Edge (f_p) & Cutoff (f_c):** The -3.01 dB point or maximum allowable passband ripple boundary A_p (dB).
- **Stopband Edge (f_s) & Attenuation (A_s):** The minimum out-of-band rejection threshold (e.g., -40 dB to -80 dB).

Step 2: Prototype Selection Strategy

Consult the decision criteria below to select the optimal approximation:

1. **Butterworth:** Recommended when monotonic passband gain and absence of ripple are paramount (e.g., measurement instrumentation, audio crossovers).
2. **Chebyshev Type I:** Recommended when sharp transition roll-off is required and passband equiripple (e.g., ≤ 0.5 dB) is acceptable.
3. **Chebyshev Type II (Inverse):** Recommended when the passband must remain strictly flat, but sharp roll-off with stopband equiripple is acceptable.
4. **Elliptic (Cauer):** Recommended when transition width must be minimized for a fixed order N , tolerating equiripple in both passband and stopband.

5. **Bessel-Thomson:** Recommended for pulse processing, biomedical signals, or audio transients where linear phase and zero group delay distortion are mandatory.

Step 3: Parameter Configuration in Overtune 3

1. Open the **Design** workspace tab (**Ctrl+1**).
2. Select the filter family from the prototype dropdown: **Butterworth**, **Chebyshev I**, **Chebyshev II**, **Elliptic**, or **Bessel**.
3. Set the filter order $N \in [1, 20]$ and cutoff frequency f_c using the interactive sliders or numeric entries.
4. For Chebyshev/Elliptic filters, configure passband ripple $R_p \in [0.1, 3.0]$ dB and stopband attenuation $A_s \in [20, 100]$ dB.

3. Interactive Analysis & Verification

3.1 Bode Frequency Response Inspection

Navigate to the **Analysis** tab (**Ctrl+2**) to evaluate frequency metrics:

- **Magnitude Plot:** Confirm passband ripple complies with A_p and the stopband achieves required rejection A_s . Hover the cursor to read exact values.
- **Phase Plot:** Examine the phase curve for discontinuities. Automatic unwrap highlights steepness of $d\theta/d\omega$ near cutoff.

3.2 Pole-Zero Constellation Analysis

Switch to the **Poles & Zeros** view (**Alt+2**):

- Verify that all poles (marked with $\times$) reside strictly inside the unit circle $|z| = 1$. The distance to $|z| = 1$ dictates resonance Q :

$$Q \approx \frac{1}{2(1 - |p|)} \quad (1)$$

Poles positioned near $|z| = 1$ produce sharp resonance peaks and high sensitivity to coefficient rounding.

- Observe transmission zeros (marked with $\circ$). Chebyshev Type II and Elliptic designs place zeros on $|z| = 1$, providing infinite attenuation notches.

3.3 Transient Impulse and Step Response

Switch to the **Impulse & Step** response plot (**Alt+3**):

- **Impulse Response $h[n]$:** Measure decay envelope and settling time. Ensure the response settles within the target sample budget without limit cycles.
- **Step Response $s[n]$:** Observe percentage overshoot $M_p = (s_{\max} - s_{\infty})/s_{\infty} \times 100\%$. Bessel filters display negligible overshoot ($< 1\%$), whereas high-order Chebyshev filters may exhibit ringing exceeding 25%.

4. Canonical Audio EQ Filter Recipes

Filter Type	Transfer Function Characteristic	Typical Audio Application
Low-Pass (LPF)	$H(0) = 1, H(\pi) = 0$	Anti-aliasing, subwoofer crossover
High-Pass (HPF)	$H(0) = 0, H(\pi) = 1$	DC blocking, vocal rumble filter
Band-Pass (BPF)	Zero at $z = \pm 1$, peak at ω_0	Instrument resonance, telecommunications
Notch (Band-Stop)	Unit zeros on $e^{\pm j\omega_0}$, $H(\omega_0) = 0$	50/60 Hz mains hum removal
Peaking EQ	Adjustable gain A at center ω_0	Parametric equalizer, vocal presence
Low/High Shelf	Transition between gain 1 and A	Bass / Treble tone controls

Table 1: Canonical audio biquad filter topologies and usage patterns.

5. Exporting Production C/C++ Code

The exported Direct Form II Transposed execution loop is organized as follows:

```
typedef struct { float b0, b1, b2, a1, a2, w1, w2; } BiquadSOS;

void process_sos_cascade(BiquadSOS* s, int count, const float* in, float* out, int N) {
    for (int n = 0; n < N; ++n) {
        float x = in[n];
        for (int k = 0; k < count; ++k) {
            float y = s[k].b0 * x + s[k].w1;
            s[k].w1 = s[k].b1 * x - s[k].a1 * y + s[k].w2;
            s[k].w2 = s[k].b2 * x - s[k].a2 * y;
            x = y;
        }
        out[n] = x;
    }
}
```

6. Laboratory Exercises

1. Synthesize an 8th-order Butterworth low-pass filter with $f_c = 1.0$ kHz at $f_s = 48$ kHz. Observe the 4 biquad sections and record the pole radii.
2. Replace the prototype with an 8th-order Chebyshev Type I filter with 0.5 dB ripple. Compare the -40 dB stopband attenuation frequency against the Butterworth design.
3. Design a 50 Hz notch filter with $Q = 30$. Export the C code and verify zero DC and Nyquist attenuation in the generated frequency response.